

RWA-PoB: A Credential-Based Proof-of-Backing Framework for Tokenized U.S. Treasury Products

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Abstract. Proof of reserves (PoR) can improve transparency for tokenized assets, but aggregate reserve coverage does not establish whether off-chain assets are legally eligible, unencumbered, consistently valued, or sufficiently liquid for redemptions. We propose RWA-PoB, a credential-based proof-of-backing framework for tokenized U.S. Treasury products. Five authorised institutional roles approve a canonical EIP-712 snapshot containing reserve, liability, liquidity, and policy information. The framework evaluates backing adequacy through the Backing Coverage Ratio (BCR) and short-term redemption capacity through the Redemption Liquidity Coverage (RLC). The Solidity prototype couples the policy controller to an ERC-20 token. Successful issuance atomically increases token supply and recorded liabilities by the corresponding USD-denominated liability. A redemption request burns tokens while reclassifying the corresponding obligation as pending. The obligation is reduced only after an authorised settlement-role account confirms payment. We evaluate the framework against a simplified aggregate PoR baseline using USDY-calibrated liabilities and deterministic synthetic reserve scenarios. Both approaches permit issuance in the valid state, but RWA-PoB rejects an encumbered-assets state with a BCR of 96.3408%, below the experimental 105% threshold. Under liquidity stress, it classifies the proposed redemption as queued because post-request RLC falls to 39.9999%. RWA-PoB authenticates the attribution and integrity of institutional claims, it does not independently prove the existence, ownership, or condition of off-chain assets. The prototype, test suite, datasets, and replication scripts are available at <https://github.com/rischanlab/PoB>.

Keywords: Proof of Backing · Real-World Asset Tokenization · Tokenized U.S. Treasuries

1 Introduction

Tokenization enables financial instruments and their associated ownership records to be represented and transferred through distributed ledgers. This architecture can improve automation and transparency in financial operations. It can also support programmable transactions, more efficient collateral use, and wider access to financial data. These benefits, however, are not automatic. Legal finality and asset custody continue to depend on institutions outside the blockchain. The same applies to valuation, redemption funding, and cash settlement.

Recent initiatives illustrate this relationship between on-chain systems and conventional financial infrastructure. BlackRock’s BUIDL is a tokenized investment fund issued on public blockchain infrastructure. Swift, UBS Asset Management, and Chainlink have demonstrated the coordination of subscriptions and redemptions for tokenized funds through existing payment systems. DTCC’s Smart NAV pilot has also examined the delivery of structured net asset value data to distributed-ledger applications [1,2,3]. Together, these initiatives demonstrate growing institutional adoption. They also show that tokenized products remain dependent on off-chain financial, operational, and legal processes.

The crypto-native concept of *proof of reserves* (PoR) is commonly used to provide evidence that an issuer or custodian holds assets corresponding to its reported obligations. Depending on its design, a PoR system may disclose wallet balances, aggregate reserve values, custodian records, or independently verified attestations. This evidence can improve transparency, but an aggregate comparison between assets and liabilities does not by itself establish solvency or liquidity. It also does not establish whether token holders have enforceable claims over the reported assets.

Vidal-Tomás distinguishes among proof of assets, proof of liabilities, proof of reserves, and proof of solvency. He argues that reserve composition and omitted obligations can materially affect the interpretation of an apparently

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sufficient reserve balance [5]. This problem is broader for real-world assets (RWAs). The existence of an off-chain asset does not necessarily establish that it is legally owned by the appropriate entity or available to token holders. The asset may also be encumbered, unsettled, incorrectly valued, or unavailable within the promised redemption period.

These limitations are particularly important for tokenized securities. As emphasised by the U.S. Securities and Exchange Commission, placing a security on a blockchain does not change the legal nature of the instrument. The same substantive legal requirements continue to apply to its on-chain representation [16]. Consequently, a technically valid token and an accurately reported on-chain supply do not independently establish the rights of token holders over the corresponding off-chain assets. A credible backing framework must therefore connect the on-chain state to evidence about custody and legal ownership. It must also account for segregation, encumbrance, valuation, settlement finality, and outstanding liabilities.

Existing systems address several parts of this problem. Chainlink Proof of Reserve can deliver off-chain or cross-chain reserve data to smart contracts. It can also be integrated with minting controls and automated circuit breakers [12]. Treasury-backed products such as Ondo USDT publish daily third-party reserve attestations. They also report underlying asset values, collateralisation levels, and portfolio maturity information [14].

The remaining problem is therefore not simply a lack of reserve disclosure or automated mint controls. Publicly documented systems do not generally define a common evidence model designed specifically for tokenized Treasury products. In particular, they do not consistently bind claims about custody, legal eligibility, encumbrance, valuation, settlement, liabilities, and redemption liquidity to the same reporting state. Without a common time reference and policy version, individually valid claims may describe different reserve conditions. This makes it difficult for a smart contract to determine whether the combined evidence supports a particular minting or redemption decision.

To address this problem, we propose *RWA-PoB*, a credential-based proof-of-backing framework for tokenized U.S. Treasury products. *RWA-PoB* is not intended to provide a trustless cryptographic proof of real-world truth. *RWA-PoB* instead provides an attributable and machine-verifiable method for evaluating institutional claims under explicit trust assumptions. These claims are issued by identified custodians, administrators, valuation agents, legal or trustee roles, and independent verifiers. Once accepted, the claims are translated into on-chain controls for minting and redemption.

The contributions of this paper are as follows:

- We define a role-separated evidence model and a canonical reserve snapshot for tokenized U.S. Treasury products. The model assigns institutional responsibilities for custody, valuation, legal eligibility, liabilities, and settlement, while binding the reported aggregates and reserve commitment to a common epoch, valuation time, and policy version.
- We distinguish backing adequacy from redemption liquidity through two policy metrics. The *Backing Coverage Ratio* (BCR) compares risk-adjusted eligible reserves with total economic liabilities, whereas the *Redemption Liquidity Coverage* (RLC) evaluates whether liquid assets are sufficient for redemption obligations under the applicable liquidity horizon.
- We implement an on-chain policy controller that verifies five required institutional signers, EIP-712 signatures, epoch progression, freshness, expiry, policy consistency, liability continuity, and replay protection. The accepted snapshot determines whether additional issuance satisfies the BCR and RLC thresholds and whether a proposed redemption receives an instant or queued disposition.
- We integrate the policy controller with a non-upgradeable ERC-20 token so that supply and liability changes occur atomically. Successful issuance increases token supply and records the corresponding USD-denominated liability. A redemption request burns tokens while reclassifying the corresponding liability as a pending-redemption obligation, which is reduced only after settlement is confirmed by an authorised role. The reported prototype evaluates the special case $P_t^{\text{red}} = 1$.
- We evaluate the prototype against a simplified aggregate PoR baseline using USDT-calibrated liabilities and deterministic synthetic reserve scenarios. The evaluation covers valid, encumbered-assets, and liquidity-stress states, together with controller tests, adversarial controller-token integration tests, a seeded accounting-invariant state machine, and local EVM gas measurements.

RWA-PoB is not intended to replace regulated custodians, administrators, auditors, or legal agreements. Its purpose is to make their relevant claims attributable, consistently structured, time-bounded, and directly usable by smart-contract policy. The framework therefore enables an on-chain system to evaluate more than the reported existence of reserve assets. It can also assess whether those assets are eligible, sufficiently liquid, and adequate to support the complete set of token-related obligations under the stated assumptions.

2 Literature Review

Proof of Reserves, Liabilities, and Solvency. Research on reserve assurance initially developed in the context of cryptocurrency exchanges and custodial platforms. Early cryptographic approaches, such as Provisions, showed that an exchange could provide privacy-preserving evidence that its controlled reserve assets exceeded the liabilities included in a committed customer-balance set [4]. Such mechanisms improve verifiability without disclosing individual balances, while related accounting research examines blockchain’s potential to support more timely and verifiable assurance processes [23]. However, cryptographic and ledger-based verification alone cannot guarantee the completeness of reported liabilities, the absence of competing claims on reserve assets, or the availability of sufficient liquidity to meet withdrawals when due.

Vidal-Tomás distinguishes among proof of assets, proof of liabilities, proof of reserves, and proof of solvency [5]. The study shows that an aggregate reserve snapshot should not be treated as a complete measure of financial resilience. Its estimated reserve buffer of approximately 6–14% depends on the assets, institutions, and stress assumptions examined and should not be applied directly to tokenized Treasury products. The broader finding is nevertheless relevant: nominal reserve coverage may be misleading when asset quality, market risk, omitted liabilities, and liquidation capacity are not considered. Chainlink similarly notes that proof of assets and proof of liabilities do not independently establish solvency, particularly when assets may be pledged elsewhere or liabilities have different priorities [6].

Lazirko *et al.* address the separation between blockchain records and conventional accounting information through their Double-Helix framework [21]. The framework distinguishes on-chain, off-chain, and intersecting transactions and argues that reserve assurance must reconcile both information environments. This is important because off-chain liabilities, expenses, investments, and transfers may affect an entity’s financial position without appearing directly on the blockchain. However, Double-Helix primary focus is the financial position of cryptocurrency exchanges and custodial platforms not the backing of a specific tokenized security. It therefore does not define how product-level claims concerning custody, legal eligibility, encumbrance, valuation, liabilities, and settlement should be assigned to responsible institutions.

RWA-PoB addresses this related but distinct problem for tokenized Treasury products. It binds role-specific institutional claims to a common signed snapshot and uses the accepted state to support minting and redemption decisions. In this respect, RWA-PoB extends integrated reserve assurance from exchange-level accounting reconciliation to product-level backing and liquidity controls.

RWA Tokenization and the On-Chain–Off-Chain State Problem. The limitations of aggregate reserve assurance become more significant when the underlying assets exist outside the blockchain. In RWA tokenization, the ledger records a digital representation whose legal and economic validity depends on external institutions and agreements. Luo *et al.* examine RWA tokenization across legal, technical, and cryptoeconomic layers [7]. Their analysis highlights the need to maintain consistency between deterministic on-chain records and externally administered assets, rights, and obligations. Taxonomic research indicates that surveyed RWA implementations commonly use hybrid architectures [8]. Within these architectures, smart contracts may govern token representation, transfers, compliance rules, pricing, and redemption workflows, whereas legal ownership, custody, dispute resolution, and reserve verification remain anchored in off-chain arrangements. Related research on blockchain-based asset trading further shows that ledger design can affect the structure and timing of settlement [22].

This separation creates related oracle and evidence problems. The oracle literature identifies off-chain data interfaces as potential failure points because they do not inherit the reliability properties of the underlying blockchain [24]. A smart contract can verify a signature, data commitment, or Merkle proof, but cannot independently determine whether a custodian holds a Treasury security, whether that position is encumbered, whether a purchase has reached settlement finality, or whether the governing agreement gives the issuer an enforceable interest in the asset. An oracle can transmit these claims on-chain, but the reliability of the resulting policy decision depends on their source, scope, timing, and governance. The design problem therefore extends beyond transmitting an aggregate reserve value to determining which institution is authorised to attest to each material fact and how related claims are bound to a common reporting state.

Credential and Token Standards. Existing standards provide several components that can support an RWA assurance architecture, although they address different parts of the system. ERC-3643 provides identity-based permissioning and compliance controls for the issuance, holding, and transfer of regulated tokens [9]. ERC-7943 defines a common interface for administrative and compliance functions in tokenized real-world assets. These functions include transfer

validation, freezing, and forced transfers [10]. Such standards support compliant token administration, but they do not define how reserve eligibility, backing adequacy, or redemption liquidity should be assessed.

The W3C Verifiable Credentials Data Model 2.0 defines a standard data model for tamper-evident credentials and presentations [11]. Information systems research similarly characterises verifiable credentials as mechanisms for machine-verifiable exchange among issuers, holders, and verifiers, subject to interoperability and governance requirements [26]. In a proof-of-backing context, a credential could represent a custodian-confirmed position, an administrator-calculated liability, or a valuation-agent price. Operational use nevertheless requires more than verifying a digital signature: the system must define authorised roles, common snapshot identifiers, validity periods, revocation procedures, and the responsibilities of each credential issuer.

Together, these standards provide useful identity, compliance, and evidence components. They do not, however, prescribe how institutional claims should be combined into a complete financial backing state.

Industry Context and Case-Study Calibration. Industry reserve-assurance systems demonstrate that reserve information can support on-chain controls. Chainlink Proof of Reserve can obtain data from off-chain custodians or other sources and publish the resulting values to smart contracts [12]. Its *Secure Mint* pattern can prevent additional issuance when reported collateral is insufficient relative to token supply [13].

Tokenized Treasury issuers also provide detailed reserve information. Ondo USDT, for example, publishes daily third-party reserve attestations covering aggregate asset values, USDT obligations, collateralisation, portfolio composition, liquidation value, and maturity indicators [14]. Its reporting methodology retains unfulfilled redemption requests within outstanding obligations until the redemption process is completed. This paper uses USDT only as the case-study product and as the calibration context for the controlled evaluation. USDT is not used as evidence of a product-specific assurance gap.

Adjacent research on money market funds and fiat-reserve-backed tokens links stability to reserve composition, redemption arrangements, and portfolio liquidity [27,29]. These findings motivate treating backing adequacy and short-horizon redemption liquidity as separate policy questions. The research gap addressed here is methodological. Although the reviewed literature and standards address reserve disclosure, oracle delivery, credential authentication, accounting reconciliation, and regulated-token controls, they treat these as distinct concerns. Taken together, they do not provide a unified specification that binds role-assigned institutional evidence to a common reporting state, evaluates backing and redemption liquidity separately, and couples the resulting decisions to atomic token issuance and redemption-accounting transitions. RWA-PoB addresses this integration problem.

Market Development and Regulatory Context. The growth of tokenized financial markets provides further motivation for stronger assurance mechanisms. Industry forecasts identify public securities and liquid collateral, including U.S. Treasury securities, as likely early drivers of institutional tokenization. Citi's 2026 *Tokenization 2030* report estimates a base-case market of approximately \$5.5 trillion in tokenized assets by 2030 [15]. This projection indicates significant market interest, but it does not imply that tokenization automatically creates liquidity [25], legal certainty, or lower financial risk. These outcomes remain dependent on the structure of the instrument, the quality of the underlying assets, the reliability of intermediaries, and the enforceability of investor rights.

In the United States, the regulatory treatment of a tokenized Treasury product depends on the rights and obligations represented by the token. Depending on its structure, the instrument may constitute a fund interest, a debt instrument, a security entitlement, or another contractual claim. SEC Commissioner Peirce has emphasised that blockchain technology does not alter the legal nature of the underlying instrument [16]. Tokenized securities therefore remain subject to applicable securities laws. An on-chain token record, reserve commitment, or smart-contract control cannot replace valid issuance documents, lawful distribution, compliant custody, and accurate disclosure [28].

A similar principle applies in the European Union. Crypto-assets that qualify as financial instruments are generally excluded from the Markets in Crypto-Assets Regulation and remain subject to the applicable financial-services framework, including MiFID II [17]. Regulation (EU) 2022/858 establishes the Distributed Ledger Technology Pilot Regime. Under this regime, authorised market infrastructures may experiment with the trading and settlement of DLT-based financial instruments under specified conditions and supervisory oversight [18]. Tokenized Treasury products cannot therefore be classified uniformly under MiCA solely because they are tokenized. Their regulatory treatment depends on their legal rights, obligations, and economic substance.

IOSCO similarly identifies implications for investor protection, market integrity, custody, settlement, interoperability, governance, and operational resilience [19]. Its technology-neutral approach confirms that the use of DLT does not remove the responsibilities normally associated with issuing, holding, transferring, valuing, and settling financial

instruments. Although these regulatory materials do not prescribe a particular proof-of-backing framework, they identify requirements that such a framework should support. These include attributable responsibility, timely disclosure, effective custody controls, operational resilience, and enforceable investor rights.

3 Comparison with Existing Systems

Table 1 positions RWA-PoB relative to three distinct approaches: document-based reserve attestations, Chainlink Proof of Reserve infrastructure, and the publicly documented USDY reserve-assurance model. These categories should not be interpreted as directly interchangeable products. A conventional attestation is principally a reporting mechanism; Chainlink Proof of Reserve is an oracle and enforcement infrastructure that can be integrated into different token systems; USDY is an operating tokenized Treasury product with its own legal, custody, verification, and disclosure arrangements; and RWA-PoB is the evidence and policy framework proposed in this paper.

Document-Based Attestations. Periodic audit or attestation reports can provide independently reviewed evidence concerning assets and liabilities at a specified reporting time. Their scope varies considerably. Some reports provide only aggregate balances, whereas others include asset categories, maturity information, custody arrangements, or reconciliation procedures. Such reports improve transparency and accountability but are normally consumed by investors, administrators, and regulators as documents. They do not inherently provide a machine-readable state that a smart contract can verify before executing an issuance or redemption action.

Chainlink Proof of Reserve. Chainlink Proof of Reserve provides infrastructure for obtaining reserve information from off-chain or cross-chain sources and publishing it to smart contracts [12]. Its *Secure Mint* pattern allows an issuing contract to compare reported reserves with token supply and to block additional issuance when collateral is insufficient. Chainlink can therefore provide both the data-delivery layer and an automated circuit-breaker mechanism.

The assurance offered by a particular Chainlink deployment nevertheless depends on the data sources, reporting methodology, oracle configuration, update policy, and smart-contract integration selected by the implementing project. Chainlink infrastructure does not, by itself, prescribe which institutional role must attest to legal ownership, custody segregation, encumbrance, settlement finality, or redemption liquidity. RWA-PoB is therefore complementary to an oracle network: a Chainlink feed or workflow could transport an RWA-PoB snapshot, while the RWA-PoB specification would define the evidence semantics, mandatory roles, financial metrics, and policy conditions applied to that snapshot.

Current USDY Reserve Assurance. USDY represents a stronger industry baseline than a simple aggregate PoR model. Ondo publicly describes USDY as debt issued by a bankruptcy-remote entity and backed by a portfolio that includes short-duration U.S. Treasury-related assets, bank deposits, and other eligible holdings under its governing arrangements [14]. Its public transparency process includes daily third-party reserve attestations, more detailed periodic reporting, aggregate underlying asset values, USDY obligations, collateralization information, liquidation value, portfolio composition, and maturity indicators.

USDY also incorporates legal and institutional protections. Public documentation identifies a verification and collateral-agent function, asset-eligibility restrictions, and a security interest intended to protect USDY holders [14]. Moreover, USDY's published reporting methodology does not remove a redemption request from outstanding obligations until the request has been fulfilled. RWA-PoB therefore does not claim to originate the accounting principle that unsettled redemptions remain liabilities.

RWA-PoB. RWA-PoB does not replace an auditor, custodian, collateral agent, oracle network, or token issuer. It defines how claims originating from these parties are transformed into one machine-verifiable backing state. The framework requires material claims to be attributable to authorized roles, bound to a common epoch and valuation cut-off, and checked for completeness, validity, freshness, revocation, and replay before they can affect token operations.

RWA-PoB further separates two financial questions that an aggregate collateralization ratio may combine or leave implicit. The Backing Coverage Ratio (BCR) evaluates whether risk-adjusted eligible reserves cover total economic obligations, whereas the Redemption Liquidity Coverage (RLC) evaluates whether sufficient assets can be converted to settlement assets within a specified redemption horizon. The accepted snapshot is then used by a smart-contract policy controller to determine whether issuance is permitted and whether a redemption can be settled immediately, queued, or subjected to predefined contingency procedures.

Table 1: Descriptive comparison of capabilities identified in publicly available sources.

Capability	Static attestation	Chainlink PoR	USDY	RWA-PoB
Independent reserve evidence	Yes	Source-dependent	Yes	Required
On-chain reserve-data delivery	No	Yes	Partial	Yes
Reserve-linked mint control	No	Yes	Not publicly shown	Yes
Role-separated claims	Report-dependent	Source-dependent	Partial	Yes
Common time-consistent snapshot	No	Implementation-dependent	Not publicly shown	Yes
Legal, custody, and encumbrance checks	Report-dependent	Source-dependent	Partial	Explicit
Pending-redemption treatment	Report-dependent	Application-dependent	Yes	Explicit
Separate solvency and liquidity metrics	Report-dependent	Application-dependent	Partial	BCR and RLC
Credential freshness and replay protection	No	Implementation-dependent	Not publicly shown	Yes
Programmatic policy enforcement	No	Supported	Partial	Yes

Table 1 provides a descriptive comparison of capabilities identifiable in publicly available sources. The comparison distinguishes between disclosure, data delivery, and policy enforcement. Chainlink provides oracle and reserve-linked control infrastructure, whereas USDY combines third-party verification, legal arrangements, and public reserve reporting. RWA-PoB contributes a common, role-separated evidence model that binds these inputs to one time-consistent snapshot and makes their acceptance an explicit condition for on-chain actions.

The comparison shows that RWA-PoB does not replace existing attestations, oracle infrastructure, or issuer controls. Its contribution is to specify how role-specific claims concerning reserves, liabilities, legal eligibility, valuation, and settlement are combined into one canonical state. The framework then verifies the completeness, freshness, and provenance of that state before applying the BCR and RLC policy conditions to minting and redemption decisions.

4 System Scope and Trust Model

This section defines the scope and operational assumptions of RWA-PoB, followed by the institutional trust relationships and threats considered in its design.

4.1 Scope and Assumptions

We consider fungible tokens representing economic claims on portfolios of short-duration U.S. Treasury securities, cash, and cash equivalents held through a fund, trust, or bankruptcy-remote special-purpose entity. The conceptual framework includes Treasury bills and notes with remaining maturities of up to two years; long-duration bonds, equities, and other RWA classes are outside scope. The reported prototype experiments use deterministically generated positions with maturities between one and 365 days.

The general framework values token obligations using an authenticated redemption price or net asset value. In this framework, token quantities are converted into USD-denominated liabilities using the redemption price included in the accepted canonical snapshot. The reported prototype evaluates the special case $P_t^{\text{red}} = 1$, under which token quantities and USD-denominated liabilities are numerically equal and no variable-price conversion is performed.

Prototype issuance is permissioned and may be initiated only by an account holding the designated issuer role. Redemption requests, however, may be initiated by any token holder with a sufficient balance. The prototype does not implement investor whitelisting, transfer restrictions, or other regulated-token compliance controls.

The framework assumes that institutional participants can provide digitally signed, time-bounded evidence. In the prototype, this evidence consists of EIP-712 signatures from five required institutional roles over the same canonical snapshot. The controller verifies signer authorisation, signature validity, timestamps, epoch progression, policy consistency, liability continuity, and replay protection. It also permits administrative revocation of the currently accepted snapshot. The prototype does not implement the complete W3C Verifiable Credentials model, a credential-status list, or an automated oracle or relayer. Such infrastructure may deliver signed snapshots on-chain but is not treated as an independent source of real-world truth.

4.2 Institutional Trust and Threat Model

The framework distinguishes five roles that approve the canonical reserve snapshot: custodian, administrator, valuation agent, legal or trustee role, and independent verifier. A separate settlement role confirms partial or final payment against

recorded redemption claims. Token issuance is initiated through an issuer role assigned to the integrated ERC-20 contract.

At the framework level, each institutional role is responsible for specified claims. For example, the custodian is responsible for holdings and custody status, the administrator for liabilities, the valuation agent for prices and haircuts, and the legal or trustee role for ownership and asset eligibility. Each of the five required signers signs the complete canonical snapshot. The Solidity contract does not independently restrict a signer to a particular subset of snapshot fields.

A valid digital signature establishes the origin and integrity of a submitted snapshot, but not the truth of its contents. RWA-PoB therefore remains dependent on institutional honesty, contractual obligations, regulatory oversight, and legal enforcement. A materially false signed statement provides attributable evidence of what an authorised account approved, but does not by itself establish fraud, legal liability, or the existence of the reported assets.

The threat model includes unauthorised or excessive issuance, omitted liabilities, asset double-pledging, stale or conflicting evidence, replayed snapshots, inconsistent reporting times, compromised signing keys, unauthorised settlement confirmation, and manipulation or interruption of snapshot delivery. Because acceptance requires one valid signature from each of the five mandatory roles over the same digest, an honest required signer can prevent acceptance of a state that it identifies as invalid. This requirement improves safety but may reduce availability when a required signer is unavailable or refuses to sign.

For the integrated token, the prototype addresses ordinary supply-accounting bypasses by requiring issuance and the corresponding liability increase to occur in one EVM transaction. A failed controller check reverts both state changes. Redemption similarly couples token burning to the reclassification of the same amount as a pending liability. These controls do not cover separately issued wrapped or bridged representations, alternative token contracts, or a malicious governance decision to authorise another liability adapter.

The prototype treats the account holding the default administrative role as trusted governance. This account can change the policy configuration, grant or revoke institutional roles, authorise token adapters, and revoke the current snapshot. Compromise or misuse of this authority remains outside the ordinary signer and token-path protections. Production deployment would therefore require additional governance controls such as multisignature approval, time-locks, restricted token registration, and monitoring of role changes.

The contract verifies authentication, signer authorisation, timestamp and epoch conditions, policy consistency, liability continuity, and replay protection. It does not independently verify the existence, ownership, completeness, valuation, or legal status of off-chain assets. Similarly, a settlement-role call records an authorised confirmation of payment but does not cryptographically prove bank finality or transfer settlement assets. Collusion among the required signers, undisclosed liabilities or legal claims, custody failures, malicious governance, and incorrect but consistently signed information remain outside the cryptographic guarantees of the framework.

5 RWA-PoB Architecture and Contribution

USDY provides a useful case study because its publicly documented model includes daily third-party reserve attestations, a redemption-price oracle, and subscription and redemption functions [14,20]. USDY provides the product context and liability-scale calibration for the controlled evaluation.

RWA-PoB is proposed as an additional assurance layer. Figure 1 distinguishes the existing assurance process from the proposed contribution.

RWA-PoB contribution. The framework contributes four linked components: **(i)** five-role snapshot approval; **(ii)** a canonical backing snapshot; **(iii)** a BCR- and RLC-based policy controller; and **(iv)** atomic token-liability integration. Together, these components convert a fresh, commonly approved backing state into enforceable issuance and redemption-accounting transitions.

5.1 Evidence and Institutional Roles

Table 2 allocates institutional responsibilities within the evidence model. Five roles approve the canonical snapshot, while a separate operational role confirms subsequent redemption settlements.

The custodian, administrator, valuation agent, legal or trustee role, and independent verifier each sign the complete canonical snapshot. The prototype therefore enforces role-separated approval, not field-level signatures. The contract

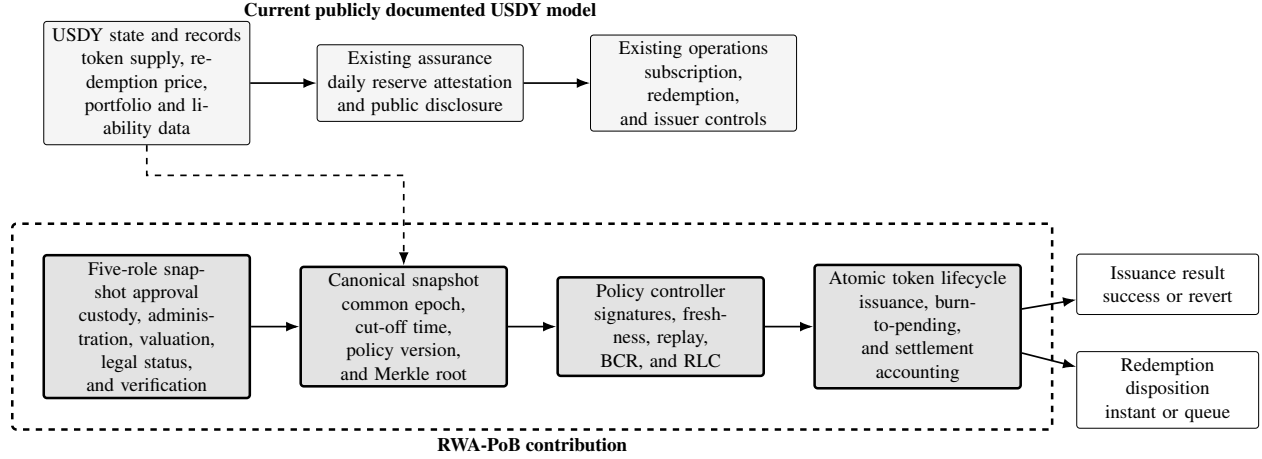


Fig. 1: Existing USDY reserve assurance and the proposed RWA-PoB extension. The contribution combines five-role snapshot approval, policy evaluation, and atomic token-liability accounting; USDY provides the case-study environment.

Table 2: Institutional roles in the RWA-PoB evidence model.

Role	Assigned responsibility	Function
Custodian	Positions, quantities, custody, encumbrance, and settlement status.	Snapshot signer
Administrator	Token liabilities, pending redemptions, fees, and other liabilities.	Snapshot signer
Valuation agent	Asset values, valuation time, haircuts, and liquidity assumptions.	Snapshot signer
Trustee or legal role	Legal ownership, asset eligibility, and applicable security interests.	Snapshot signer
Independent verifier	Reconciliation of reserve, liability, and supporting records.	Snapshot signer
Redemption agent	Confirmation of partial or final payment against a redemption claim.	Settlement role

verifies exactly one authorised signer for each required role, while the allocation of responsibility in Table 2 remains an off-chain institutional requirement.

The settlement role is not a required snapshot signer. It confirms partial or final payment against an existing redemption identifier. This logical separation must be maintained through role assignment because the contract does not prevent governance from assigning settlement and snapshot roles to the same account. The token’s issuer role is also separate from the administrator role and is used only to initiate atomic issuance.

5.2 Reserve and Liability Model

Eligible Reserves The general framework represents each direct reserve position j as

$$RSV_{j,t} = \langle id_j, q_{j,t}, p_{j,t}, \tau_j, \kappa_{j,t}, \omega_{j,t}, \eta_{j,t}, \sigma_{j,t}, d_{j,t} \rangle, \quad (1)$$

where id_j is the unique identifier of reserve position j , $q_{j,t}$ and $p_{j,t}$ denote its quantity and price, τ_j is its maturity date, $\kappa_{j,t}$ identifies the custody account, $\omega_{j,t}$ represents legal ownership, $\eta_{j,t}$ records encumbrance, $\sigma_{j,t}$ records settlement status, and $d_{j,t}$ is the estimated cash-conversion horizon.

A position is eligible only when the required legal and operational conditions are satisfied:

$$\alpha_{j,t} = \mathbf{1} \left\{ \begin{array}{l} \text{custody confirmed, legal title valid,} \\ \text{no prohibited encumbrance, and settlement final} \end{array} \right\}. \quad (2)$$

Its eligible value is

$$V_{j,t}^{\text{eligible}} = \alpha_{j,t}(1 - h_{j,t})p_{j,t}q_{j,t}, \quad (3)$$

where $h_{j,t}$ is an asset- and horizon-specific haircut. The reported experiment uses simplified synthetic records containing market, eligible, and liquid values, maturity, haircut, encumbrance, legal eligibility, and settlement-finality fields. Position eligibility and aggregation are computed off-chain.

Total eligible reserves are

$$A_t^{\text{eligible}} = A_t^{\text{cash}} + \sum_j V_{j,t}^{\text{eligible}}. \quad (4)$$

A Merkle root commits to the position records but does not prove their completeness, truth, or consistency with the reported aggregates. The five required signers approve the same root and aggregate values. On-chain validation checks that the root is non-zero and that liquid assets do not exceed eligible reserves, which do not exceed gross reserves.

Economic Liabilities For a multi-network product, global economic supply is

$$S_t^{\text{econ}} = \sum_{c \in \mathcal{C}} S_{c,t} - B_t^{\text{locked}}, \quad (5)$$

where B_t^{locked} denotes tokens locked to support wrapped or bridged representations included elsewhere in the summation. In-transit amounts are treated under an explicit cross-chain policy. The prototype models one token contract, for which $B_t^{\text{locked}} = 0$.

Token obligations are valued using the authenticated redemption price included in the accepted canonical snapshot:

$$V_t^{\text{tokens}} = S_t^{\text{econ}} P_t^{\text{red}}. \quad (6)$$

This formulation captures the economic logic of an accumulating token such as USDT, for which the redemption price may change over time. The reported prototype and experiments evaluate the special case $P_t^{\text{red}} = 1$. Consequently, token quantities and USD-denominated liabilities are numerically equal in the reported implementation, although they are distinct quantities in the general framework.

Under the prototype's burn-first redemption model, total liabilities are

$$L_t = V_t^{\text{tokens}} + V_t^{\text{pending}} + O_t, \quad (7)$$

where O_t includes fees and other liabilities, and V_t^{pending} contains burned token claims that have not settled. A lock-based implementation would need to avoid counting the same obligation in both circulating supply and pending claims.

For a redemption request of q token units, the corresponding liability is qP_t^{red} , using the authenticated redemption price when the request is accepted. The request reduces V_t^{tokens} and increases V_t^{pending} by the same USD-denominated amount, leaving total liabilities unchanged.

When an authorised settlement-role account confirms payment of an amount s , the prototype assumes that settlement is made from eligible cash. The corresponding transition is

$$\begin{aligned} V_{t+1}^{\text{pending}} &= V_t^{\text{pending}} - s, \\ A_{t+1}^{\text{cash}} &= A_t^{\text{cash}} - s, \\ A_{t+1}^{\text{liquid}} &= A_t^{\text{liquid}} - s, \\ A_{t+1}^{\text{eligible}} &= A_t^{\text{eligible}} - s, \\ A_{t+1}^{\text{gross}} &= A_t^{\text{gross}} - s, \\ L_{t+1} &= L_t - s. \end{aligned} \quad (8)$$

These reductions are recorded as post-snapshot adjustments so that the original signed snapshot remains unchanged. The settlement-role call records an institutional confirmation of payment; it does not independently prove bank finality.

Equation 8 describes the complete framework-level accounting transition. The reported prototype implements only the reductions in pending liabilities and tracked liquid assets. It does not propagate settlement into cash, eligible reserves, or gross reserves between snapshots.

5.3 Backing Metrics and Policy Logic

RWA-PoB separates backing adequacy from short-horizon liquidity:

$$BCR_t = \frac{A_t^{\text{eligible}}}{L_t}, \quad RLC_{t,h} = \frac{A_{t,h}^{\text{liquid}}}{R_{t,h}^{\text{due}}}, \quad (9)$$

where $A_{t,h}^{\text{liquid}}$ is the eligible value convertible to the settlement asset within horizon h , and $R_{t,h}^{\text{due}}$ is the redemption value due within that horizon. The metrics use the effective values after post-snapshot issuance, redemption, and settlement adjustments. The prototype uses one implicit horizon: current RLC divides liquid assets by pending redemptions, while post-request RLC includes the proposed redemption in the denominator. When no redemption value is due, the RLC condition is satisfied by definition.

Snapshot acceptance requires

$$\text{Accept}(\Sigma_t) = \text{Roles} \wedge \text{Signatures} \wedge \text{Freshness} \wedge \text{Epoch} \wedge \text{Policy} \wedge \text{Values} \wedge \text{Continuity} \wedge \text{UnusedDigest}. \quad (10)$$

Governance may subsequently revoke the accepted snapshot. A revoked, expired, or policy-inconsistent snapshot is not fresh and cannot support issuance.

The policy controller applies the following rules:

1. **Snapshot update.** A snapshot is accepted only when one authorised signer from each mandatory role signs the same EIP-712 digest, its epoch increases, its timestamps and values are valid, its token liability reconciles with the economic supply and authenticated redemption price, its pending liability reconciles with unpaid redemption claims, and its digest has not been used.
2. **Minting.** The prototype assumes that subscription funds have settled and are already included in eligible reserves. For a positive issuance of q token units, let $\Delta V_t(q) = qP_t^{\text{red}}$ denote the corresponding USD-denominated liability using the authenticated redemption price. Minting is permitted only when the snapshot is fresh and

$$BCR_t^{\text{post}}(q) = \frac{A_t^{\text{eligible}}}{L_t + \Delta V_t(q)} \geq \theta_S \quad \text{and} \quad RLC_{t,h} \geq \theta_L. \quad (11)$$

Approval, the increase in token supply by q , and the increase in recorded liabilities by $\Delta V_t(q)$ occur atomically.

3. **Redemption initiation.** For a request of q token units, the integrated token burns q tokens and atomically moves $\Delta V_t(q)$ from token liabilities to pending-redemption liabilities. The applicable redemption price is fixed when the request is accepted. If the controller rejects the request, the burn is reverted.
4. **Redemption disposition.** A request receives an instant disposition only when the snapshot is fresh and post-request RLC, including $\Delta V_t(q)$ in the denominator, meets the threshold. Otherwise, it receives a queue disposition. These values classify the request but do not transfer assets or implement queue ordering.
5. **Settlement.** An authorised settlement-role account confirms a positive partial or final amount s that does not exceed the unpaid claim or available liquid assets. The reported prototype reduces pending liabilities and tracked liquid assets by s and closes the claim when fully settled. A complete implementation should also update the cash, eligible-reserve, and gross-reserve aggregates from which payment is made.

A low BCR blocks further issuance, whereas a low RLC changes the redemption disposition without extinguishing the investor's claim. The implemented flows are shown in Appendix A.

6 Prototype and Evaluation

6.1 Prototype Implementation

The prototype was implemented in Solidity 0.8.28 and evaluated using Hardhat 3.11.1 on a local Ethereum-compatible network. It comprises three deployed contracts and a narrow interface. The `PoBPolicyController` validates snapshots, calculates BCR and RLC, authorises liability changes, records redemption dispositions, and accounts for settlement. The reported implementation evaluates the special case $P_t^{\text{red}} = 1$, so token quantities and USD-denominated liabilities are numerically equal. Variable-price conversion and conversion rounding are not implemented.

Table 3: Observed decisions in the USDY-calibrated scenarios.

Scenario	Aggregate PoR mint	RWA-PoB mint	Disposition	BCR	Post-request RLC
Valid state	Accept	Accept	Instant	114.1209%	199.9999%
Encumbered assets	Accept	Reject	Instant	96.3408%	199.9999%
Liquidity stress	Accept	Accept	Queue	114.1209%	39.9999%

The `RWABackedToken` atomically couples ERC-20 issuance and burning to the corresponding liability changes in the controller. The `AggregatePoRBaseline` provides the gross-reserve comparison.

Each canonical snapshot is signed using EIP-712 by the custodian, administrator, valuation agent, legal or trustee role, and independent verifier. The controller checks signer roles and signatures, epoch and time conditions, policy consistency, value bounds, liability reconciliation, and replay protection. Under $P_t^{\text{red}} = 1$, token liabilities reconcile directly with the registered token supply, while pending liabilities reconcile with unpaid redemption claims. An accepted snapshot may subsequently be revoked. Following confirmed settlement, the controller reduces effective pending liabilities and liquid assets by the confirmed USD-denominated amount. Because total liabilities are computed from token liabilities, pending liabilities, and other liabilities, this reduction also decreases total liabilities. The prototype does not update cash, eligible reserves, or gross reserves following settlement.

The implementation, tests, scenarios, and reproduction scripts are available at <https://github.com/risch-anlab/PoB>.

6.2 Experimental Design

The evaluation tests the policy layer and does not audit or reconstruct USDY’s reserve portfolio. Liability scale is calibrated using the `RWA.xyz Bridged Token Value (Dollar)` series for Ondo U.S. Dollar Yield [30]. The processed dataset contains 1,043 daily observations from 18 September 2023 to 26 July 2026. The latest observation, approximately USD 2.162 billion, is used in the experiments.

Synthetic position records contain market, eligible, and liquid values, maturity, haircut, encumbrance, legal-eligibility, and settlement-finality fields. Each scenario contains 100 positions generated with seed 42. Gross reserves equal 115% of observed token value, other liabilities equal 0.5%, and the proposed redemption equals 5%. The experimental thresholds are $\theta_S = 105\%$ for BCR and $\theta_L = 100\%$ for RLC.

Three scenarios are evaluated:

1. **Valid state:** eligible reserves exceed the BCR threshold and liquid assets equal 10% of observed token value.
2. **Encumbered-assets state:** gross reserves remain unchanged, but encumbered positions are excluded from eligible reserves.
3. **Liquidity-stress state:** eligible reserves remain sufficient, but liquid assets fall to 2% of observed token value.

The complete suite contains 31 deterministic tests: seven controller unit tests, 21 atomic integration and adversarial tests, one seeded model-based test of 128 state transitions, and two scenario and gas-generation tests. The suite covers signature and snapshot validation, policy enforcement, role bypasses, atomic rollback, redemption and settlement bounds, and accounting invariants. All 31 tests passed in the post-merge continuous-integration run.

The USDY data are used to calibrate the dollar value of token liabilities. The experiments do not reproduce USDY’s historical token supply or changing redemption price. Instead, they set $P_t^{\text{red}} = 1$ to isolate the effects of reserve eligibility, backing coverage, redemption liquidity, and liability reclassification.

6.3 Experimental Results

Table 3 reports the observed policy decisions. The aggregate PoR baseline permitted minting in all three scenarios because gross reserves exceeded token obligations. RWA-PoB agreed in the valid state but rejected minting when encumbrance reduced eligible backing below the threshold.

In the encumbered-assets scenario, gross reserves were approximately USD 2.486 billion against token obligations of approximately USD 2.162 billion. The aggregate baseline accepted minting, whereas RWA-PoB produced a BCR of 96.3408%, below the 105% threshold, after excluding encumbered assets and including other liabilities.

Table 4: Local EVM gas consumption for reserve-state publication.

State publication	RWA-PoB	Aggregate PoR
Initial valid state	362,241	94,689
Encumbered-assets update	194,904	37,789
Liquidity-stress update	197,716	37,789

In the liquidity-stress scenario, BCR remained 114.1209%, but liquid assets covered approximately 40% of the proposed redemption. The controller therefore returned a queue disposition. This scenario test evaluates the disposition without submitting a redemption; separate integration tests confirm that a submitted request remains a pending liability until settlement.

6.4 Execution Cost

Gas usage was obtained from local Hardhat transaction receipts. Table 4 separates initial publication from subsequent updates because the first transaction initializes empty storage.

Initial RWA-PoB publication consumed approximately 3.83 times the baseline cost, while subsequent updates consumed approximately 5.16–5.23 times the baseline. The difference reflects five-signature verification and additional state storage.

In ten controlled repetitions, atomic issuance used 132,573 gas. Separate advisory authorisation and independent ERC-20 minting used 82,172 and 70,834 gas respectively, or 153,006 gas across two transactions. Atomic issuance was therefore 20,433 gas, or 13.35%, lower in this experiment while enforcing one synchronised state transition.

Scenario inputs, Merkle roots, and policy decisions reproduce deterministically under the recorded dependencies and seed. Gas values are run-specific because timestamp-dependent EIP-712 signatures can change calldata composition. They exclude deployment, off-chain processing, oracle fees, and network prices.

7 Regulatory and Legal Considerations

The legal treatment of a tokenized Treasury product depends on its structure and the rights represented by the token. In the United States, it may constitute a fund interest, debt security, security entitlement, or other contractual claim. Distributed-ledger technology does not change the legal character of an underlying security [16]. Applicable requirements may include registration or exemptions, disclosure, custody, transfer restrictions, investor eligibility, and anti-money-laundering controls.

In the European Union, MiCA does not apply to crypto-assets that qualify as financial instruments, which remain subject to the applicable securities-market framework. The DLT Pilot Regime establishes a supervised framework for certain DLT-based market infrastructures [17,18]. Product classification and jurisdiction must therefore be determined before selecting legal, credential, and policy requirements.

RWA-PoB does not determine or enforce legal compliance. The conceptual framework could be combined with ERC-3643 or external identity and compliance systems [9]. The implemented token, however, is a standard ERC-20 with unrestricted transfers. It does not implement investor whitelisting, identity verification, freezing, forced transfers, jurisdictional restrictions, or anti-money-laundering controls.

A document hash or Merkle commitment establishes that presented data match a commitment; it does not prove the data are true or create legal rights. Enforceable rights must arise from applicable fund documents, security agreements, custody arrangements, and law. In the framework, a position whose ownership, custody, or enforceability cannot be established receives $\alpha_{j,t} = 0$ and is excluded from eligible reserves. The prototype does not establish these legal facts independently; it relies on off-chain eligibility classification and institutional approval of the signed snapshot.

8 Limitations and Future Research

RWA-PoB authenticates the origin and integrity of signed institutional statements, not their economic or legal truth. Signer collusion, omitted liabilities, false custody or valuation records, and defective legal arrangements may therefore

remain undetected. The prototype enforces role-authorised approval but does not independently establish organisational independence among the signers or restrict each signer to specific snapshot fields. A settlement-role call records an institutional confirmation of payment but does not prove payment or bank finality. Similarly, a Merkle root commits to position records but does not establish portfolio completeness or consistency with the reported aggregates, and the controller neither verifies position proofs nor recomputes reserve totals. Governance can change policies, grant or revoke roles, authorise token contracts, and revoke the current snapshot. Requiring all five roles to approve each snapshot improves safety but creates an availability risk because one unavailable or withholding signer can prevent an update, halt issuance, and cause redemptions to be queued.

The general framework supports a snapshot-authenticated redemption price, whereas the reported prototype evaluates the special case $P_t^{\text{red}} = 1$. The simplified PoR baseline does not represent encumbrance or redemption liquidity, so the comparison illustrates the effect of applying the additional eligibility and liquidity conditions. The experiments use the latest observation from the 1,043-day liability series, synthetic positions generated with seed 42, one deterministic state-machine sequence, and local EVM gas measurements. They do not provide longitudinal simulation, real portfolio calibration, valuation or availability sensitivity, scaled network-cost analysis, formal verification, an independent security audit, or production validation.

Future work should formalise the adversary model and the safety and liveness properties of the framework, including supply-liability reconciliation under price updates, liability conservation during redemption initiation, and reserve reduction following settlement. These properties should be evaluated through fuzz-invariant testing, symbolic execution, and a threat-to-outcome matrix. Longitudinal simulations should use the complete liability series, stochastic redemption flows, and parameter sweeps to measure threshold binding, queue duration, false pauses, and valuation sensitivity. Further evaluation should use position-level regulatory or issuer data and compare RWA-PoB with aggregate-reserve, eligible-reserve, and single-attestor baselines. Future work should also quantify signer availability, examine threshold-signature alternatives, calibrate policy parameters against applicable requirements, and measure gas across signer counts, policy paths, and deployment networks. Production studies should integrate live price and credential-status infrastructure, verifiable settlement evidence, compliant transfers, cross-chain supply reconciliation, and hardened governance.

9 Conclusion

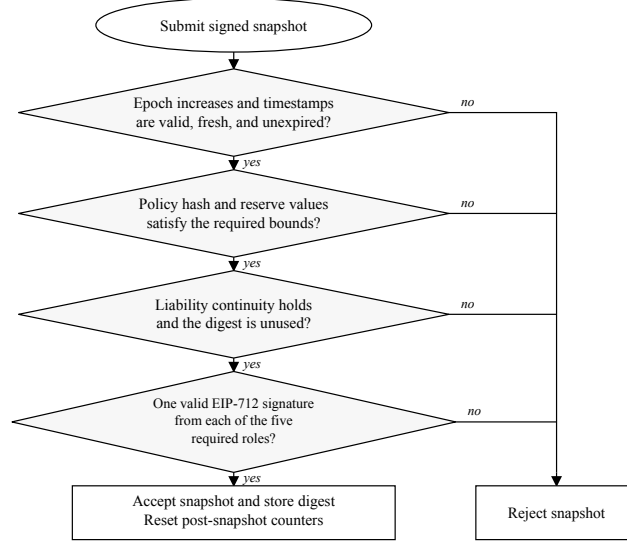
Aggregate proof of reserves does not by itself establish that reported assets are legally eligible, unencumbered, consistently valued, or available within a required redemption horizon. This paper proposes RWA-PoB, a credential-based assurance framework for tokenized U.S. Treasury products. It combines five-role approval of a canonical snapshot with explicit backing, liquidity, issuance, and redemption-accounting policies.

RWA-PoB makes four principal contributions. First, it verifies signer roles, signatures, freshness, policy consistency, liability reconciliation, and replay protection. Second, it separates backing adequacy from redemption liquidity through BCR and RLC. Third, it defines the atomic coupling of ERC-20 issuance to the corresponding USD-denominated liability. The reported prototype evaluates this mechanism under the special case $P_t^{\text{red}} = 1$. Redemption requests reclassify the same USD-denominated value as a pending obligation until settlement is confirmed. Fourth, it provides reproducible scenario, invariant, adversarial, and gas evidence for these controls.

In the controlled evaluation, RWA-PoB rejected an encumbered-assets state that the aggregate PoR baseline accepted. Under liquidity stress, it classified the proposed redemption as queued despite adequate overall backing. Integration tests further showed that submitted redemptions remain pending liabilities until settlement and that confirmed payments reduce both the outstanding liability and tracked liquid assets. These results demonstrate the intended policy and accounting behaviour of the prototype but do not remove institutional trust or establish production readiness. Broader empirical, legal, operational, and security validation remains necessary.

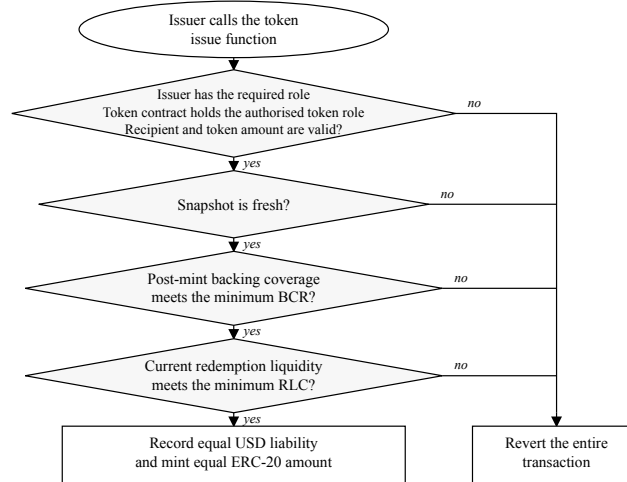
A RWA-PoB Policy-Flow Diagrams

This appendix presents the operational decision flows implemented by the RWA-PoB prototype. Figure A.1 summarises snapshot validation and mint authorisation. Figure A.2 shows redemption classification and subsequent settlement confirmation.

(a) Snapshot validation

All five roles approve the same EIP-712 snapshot digest.

The prototype fixes the redemption price at one. Variable-price validation is outside this flow.

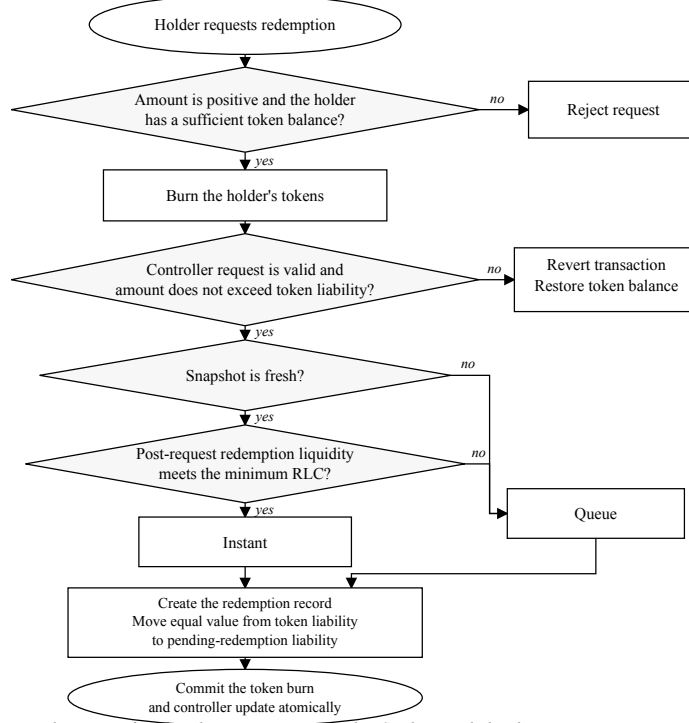
(b) Atomic mint authorisation and issuance

With a unit redemption price, the liability increase equals the token amount.

The liability update and token mint commit atomically.

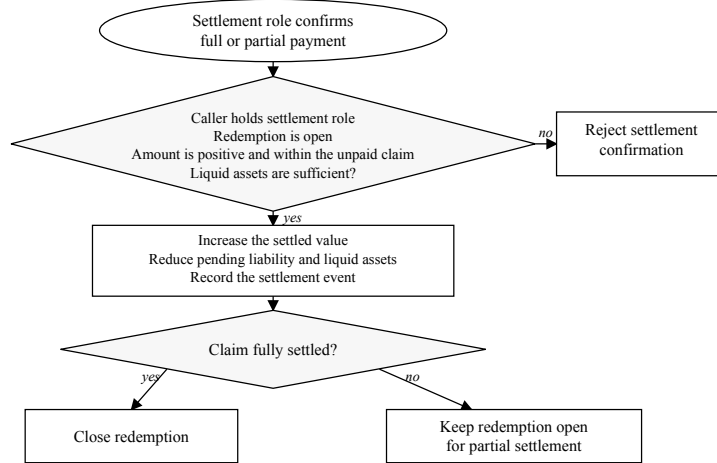
Fig. A.1: RWA-PoB snapshot-validation and mint-authorisation flows.

(a) Atomic redemption classification



*The prototype fixes the redemption price at one, so the USD claim equals the token amount.
Classification does not transfer assets. Total liabilities remain unchanged.*

(b) Settlement confirmation



*Prototype settlement updates pending liabilities and tracked liquid assets but does not transfer cash.
The complete framework should also update cash, eligible reserves, and gross reserves.*

Fig. A.2: RWA-PoB redemption-classification and settlement-confirmation flows. A redemption burns q tokens while reclassifying the corresponding request-time USD liability $\Delta V_t(q)$ as pending. In the reported prototype, $P_t^{\text{red}} = 1$ and therefore $\Delta V_t(q) = q$.

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